

Measuring the Ceiling: Evidence That Easy-to-Medium Cybersecurity MCQ Benchmarks Have Saturated for Current-Generation LLMs

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Abstract—Cybersecurity benchmarks that rely on multiple-choice questions (MCQ) were designed to measure and rank large language model capabilities. But what happens when the models outpace the measurement instrument? This paper provides reproducible, live-API evidence that easy-to-medium difficulty cybersecurity MCQs—the kind that form the bulk of most existing benchmark question banks—have reached saturation for current-generation models. We evaluate eight frontier and near-frontier large language models across 210 cybersecurity questions spanning seven security dimensions. Seven of eight models score at or above 91.0%; three achieve perfect scores. However, the question bank is concentrated in difficulty tiers 1–3 (200 of 210 items), meaning the saturation finding applies to the easier end of the difficulty spectrum. At tier 4, ceiling effects persist for the strongest models but discrimination remains possible. We further document a 39.5 percentage-point gap between scores predicted by deterministic simulation and live measurements, reflecting model capability improvement over approximately 18 months. Based on these findings, we argue that: (1) existing easy-to-medium cybersecurity MCQ benchmarks should be explicitly reframed as floor tests—they can flag unsuitable models but cannot rank competent ones; (2) benchmark saturation should be treated as a publishable research output that tracks the movement of the evaluation frontier; and (3) the hard-question tail must be substantially expanded before claims about MCQ format discrimination can be evaluated at the frontier. We release the full evaluation pipeline, question bank, response traces, and scoring logic as a reproducible artifact.

Index Terms—LLM evaluation, cybersecurity benchmarks, benchmark saturation, multiple-choice questions, empirical measurement, model comparison.

I. INTRODUCTION

THE deployment of large language models (LLMs) in cybersecurity is accelerating across vulnerability triage, threat intelligence analysis, secure code review, incident response, and governance support [16]. Since the transformer architecture [47] enabled the scaling of language models [48], [49], [50], capabilities have expanded rapidly, with emergent behaviors appearing at scale [51]—though recent work questions whether emergence is a measurement artifact [57]. Foundation models [25] now underpin systems from general-purpose assistants [27], [28], [29] to specialized coding agents [32], [33], driven in part by chain-of-thought reasoning [30] and zero-shot generalization [31]. With this acceleration comes

an acute evaluation problem: organizations deploying LLMs for security workflows need reliable evidence about model suitability, not aggregated leaderboard rankings that obscure domain-specific strengths and weaknesses.

Current cybersecurity evaluation practice relies heavily on multiple-choice question (MCQ) benchmarks. CyberSecEval [8], [9], [10], CyberMetric [11], CTIBench [12], and SecBench [15] all use MCQ formats as primary or supplementary evaluation channels. The MCQ format is attractive: it is objective, reproducible, and easy to scale. But its utility depends on an assumption that is rarely examined directly—that the format continues to discriminate between models as capabilities improve.

This paper tests that assumption empirically and finds that, for easy-to-medium questions, it is failing. We evaluate eight models from five providers against a 210-item cybersecurity question bank spanning seven dimensions. The result: seven of eight models score at or above 91%. Three score perfectly.

The contribution is a measurement of where the evaluation frontier currently stands for the most common type of benchmark item. Prior work on benchmark lifecycles has established that saturation is inevitable [4], [5]. Our contribution is empirical: we establish *that* saturation has occurred for easy-to-medium cybersecurity MCQs and document the magnitude of the capability improvement. An important scope limitation is that our question bank contains only 10 items at difficulty tiers 4–5, so the claim is specific to the easier end of the spectrum. Our contributions are:

- 1) **Reproducible measurement of easy-to-medium MCQ saturation.** Live-API measurements across eight models and 210 questions show seven of eight models scoring at or above 91.0%. The easy-to-medium MCQ format no longer discriminates between current-generation models.
- 2) **Documentation of the capability improvement trajectory.** The 39.5 percentage-point gap between prior-generation benchmark predictions and our current-generation measurements quantifies the pace of cybersecurity knowledge improvement in LLMs over approximately 18 months.
- 3) **A reframing of the role of MCQ benchmarks.** Easy-to-medium MCQ benchmarks should transition from ranking

tools to floor tests: a model scoring below 90% on such questions is clearly unsuitable for deployment, but scores above this threshold provide no further discrimination.

- 4) **Evidence-based scope recommendations.** We identify the hard-question tail as the critical missing component in current MCQ evaluation and recommend that future benchmarks allocate at least 50% of items to difficulty tiers 4–5.
- 5) **A fully reproducible artifact.** The evaluation pipeline, question bank, response traces, and scoring logic are released as a single-executable artifact.

The remainder of the paper is organized as follows. Section II surveys related work across LLM evaluation, cybersecurity benchmarking, and benchmark saturation. Section III describes the CyberEval framework design and evaluation protocol. Section IV presents the experimental methodology. Section V reports the results. Section VI discusses implications for evaluation practice, benchmark design, and deployment. Section VII addresses limitations. Section VIII concludes.

II. RELATED WORK

A. General LLM Evaluation

The evaluation of large language models has matured rapidly since the release of foundational frameworks such as HELM [1], which established the principle of multi-axis measurement. Survey work by Chang et al. [2] synthesizes a growing design space and identifies format sensitivity—the observation that benchmark conclusions depend on prompt design, task construction, and rubric choice—as a central challenge. MMLU [26] pioneered broad-coverage knowledge evaluation across 57 subjects, establishing a paradigm now widely adopted. More recently, Polo et al. [3] demonstrated that benchmark efficiency and calibration matter as much as raw scale, especially when evaluation must be repeated or updated.

Two developments in the broader evaluation literature are particularly relevant to this paper. First, the recognition that benchmarks have lifecycles: they are created, they measure, they saturate, and they are replaced [4]. Second, the finding that evaluation format can reverse model rankings: models that appear equivalent under one format may diverge sharply under another [1], [2]. Ganguli et al. [56] provide evidence that capability gains can outpace benchmark design cycles. Lin et al. [54] extend holistic evaluation to agentic deployments. Both insights are central to our argument.

B. Cybersecurity LLM Benchmarks

Cybersecurity benchmark development has accelerated since 2023, driven by the dual pressures of identifying risky capabilities and evaluating beneficial deployment suitability.

Secure coding benchmarks. The CyberSecEval line [8], [9], [10] is the most prominent secure coding evaluation effort, initially focusing on insecure code generation detection and later expanding to broader cyber-risk assessment including vulnerability exploitation and attack safety. Pearce et al. [17] and Perry et al. [18] provided early evidence that LLM-assisted

coding could introduce security vulnerabilities and inflate user confidence.

Knowledge-oriented benchmarks. CyberMetric [11] evaluates cybersecurity knowledge using retrieval-augmented benchmark construction. CTIBench [12] focuses specifically on cyber threat intelligence assessment. Both adopt MCQ as a primary evaluation format and report scores in the 40–70% range for models evaluated in that generation.

Offensive and agentic benchmarks. NYU CTF Bench [13] introduced a scalable dataset for evaluating LLMs on offensive security tasks derived from capture-the-flag competitions. CyBench [14] provides a broader framework for evaluating both cybersecurity capabilities and associated risks. Complementary work on LLM penetration testing [19], [20], [37], [38] and automated security agents [36] extends evaluation toward task execution. SWE-bench [34] and SWE-agent [35] provide general software engineering evaluation frameworks whose methodology informs cybersecurity agent assessment. PentestBench [46] and SecureVal [39] offer focused evaluation for offensive security and defensive security capabilities respectively.

Broad-coverage benchmarks. SecBench [15] represents the most comprehensive recent effort, spanning multiple cybersecurity domains. The systematic review by Zhang et al. [16] documents the rapid expansion of the cyber-LLM landscape across vulnerability analysis, threat intelligence, malware analysis, secure development, and red teaming. Complementary surveys by Chen et al. [43] and Wang et al. [44] provide comprehensive taxonomies of LLM security capabilities and evaluation approaches. Liu et al. [45] identify specific gaps between benchmark design and security operations requirements.

Safety and dual-use benchmarks. Beyond capability evaluation, a parallel literature addresses safety risks. WMDP [21] measures hazardous knowledge, HarmBench [22] provides standardized red-teaming evaluation, R-Judge [23] assesses safety risk awareness in agentic settings, and WildTeaming [24] scales adversarial safety evaluation. These works, while not directly comparable to our MCQ saturation study, share our concern that evaluation design shapes conclusions about model behavior.

C. Benchmark Saturation

Benchmark saturation is a well-documented phenomenon in machine learning [4], [5], [53]. Benchmarks are created to measure a capability; models improve; scores converge toward ceiling performance; and the benchmark loses discriminatory power. The standard response is to build a harder benchmark [6], [7]. Bender et al. [52] and Kapoor et al. [55] have cautioned that benchmark-driven evaluation, when decoupled from deployment context, can produce misleading conclusions about model readiness. Our work extends this critique to cybersecurity evaluation specifically.

Our contribution to this literature is specific but important: we provide reproducible evidence that *this specific* transition has occurred for cybersecurity MCQ evaluation. The saturation is recent. Benchmarks published two years prior

reported scores in the 40–70% range [11], [12], [13], [15]. Our measurements, conducted at time of writing, return 91–100%. The speed of this transition—roughly 30–50 percentage points in under two years—is itself a signal about the pace of cybersecurity LLM capability improvement and the fragility of static evaluation instruments.

D. Positioning of This Work

This paper is distinguished from prior work in three respects. **First**, unlike simulation-based studies, we provide live-API evidence from eight models across five providers. **Second**, unlike benchmark proposals that treat saturation as a limitation to be addressed in future work, we treat saturation as the primary finding and analyze its implications for evaluation design. **Third**, unlike work that treats MCQ evaluation as an unalloyed good, we provide evidence that easy-to-medium MCQ benchmarks have reached their useful limit for ranking current-generation models.

III. CYBEREVAL FRAMEWORK DESIGN

A. Security Dimensions

CyberEval organizes cybersecurity competence into seven dimensions derived from recurring security workflows in both academic benchmark literature and practitioner deployment contexts:

- 1) **Vulnerability Knowledge (VK)**: Recognition and classification of software and system vulnerabilities, including CWE taxonomy, CVSS scoring, and exploit mechanics.
- 2) **Threat Intelligence (TI)**: Understanding of threat actors, attack campaigns, indicators of compromise, and the intelligence lifecycle.
- 3) **Secure Coding (SC)**: Knowledge of secure development practices, common coding vulnerabilities, and remediation patterns.
- 4) **Incident Response (IR)**: Understanding of detection, containment, eradication, and recovery workflows, including NIST and SANS frameworks.
- 5) **Compliance & Governance (CG)**: Knowledge of regulatory frameworks (GDPR, NIST 800-53, ISO 27001), risk management, and policy design.
- 6) **Forensic Analysis (FA)**: Understanding of digital evidence collection, chain of custody, memory and disk forensics, and timeline reconstruction.
- 7) **Security Architecture (SA)**: Knowledge of network segmentation, zero-trust design, cloud security models, and cryptographic protocol selection.

The dimensions are pragmatic rather than ontological. They correspond to distinct hiring profiles, certification domains, and operational workflows in security organizations, which makes them directly relevant to deployment-oriented model selection.

B. Question Bank

The evaluation uses a 210-item seed bank with 30 questions per dimension. Each question presents four answer choices

with exactly one correct answer. Questions carry a heuristic difficulty annotation in the range 1–5, assigned during construction based on expected depth of domain knowledge required.

The bank is concentrated in difficulty tiers 1–3, with 37 items at tier 1, 116 at tier 2, 47 at tier 3, 9 at tier 4, and only 1 at tier 5. This imbalance—inherited from the original framework design, which prioritized broad coverage over calibrated difficulty scaling—is a central limitation of our study and constrains the scope of our claims to the easy-to-medium difficulty range.

Example questions. Table I provides illustrative items from three difficulty tiers. These examples demonstrate the style and domain coverage of the question bank; the full set of 210 questions is included in the released artifact.

Question construction. Questions were authored by cybersecurity researchers and practitioners affiliated with the VIA Cybersecurity Lab at KNUST. The seven-dimension taxonomy was derived from recurring security workflows in both academic benchmark literature and practitioner deployment contexts. Each item was assigned a heuristic difficulty rating (1–5) based on the depth of domain knowledge and the number of reasoning steps required. Items at tier 1 test recall of widely known facts; items at tier 3 require synthesis of multiple concepts; items at tier 5 require expert-level reasoning about edge cases. The distribution reflects an initial design goal of broad coverage across dimensions rather than calibrated difficulty scaling—a choice that, in retrospect, limits the current study’s ability to characterize hard-question discrimination.

C. Prompt Template

Each question is presented to every model using a standardized template designed to minimize format-induced variance:

“Cybersecurity MCQ. Choose the BEST answer. Reply with ONLY the letter (A, B, C, or D).

Question: [question text]

Options:

A. [choice text]

B. [choice text]

C. [choice text]

D. [choice text]

Your answer (single letter):”

The template explicitly instructs models to respond with a single letter. For measuring MCQ saturation, a single-letter response protocol is sufficient and produces unambiguous, auditable results.

D. Answer Extraction and Scoring

Responses are parsed by extracting the last occurrence of a single uppercase letter (A, B, C, or D) in the model output. If no such letter is found, the response is scored as incorrect. This conservative strategy ensures that models are credited only when they produce an unambiguous answer. It may slightly underestimate model knowledge in cases where the model produces the correct answer in an unconventional format, but this bias is consistent across all models and therefore does not distort relative comparisons.

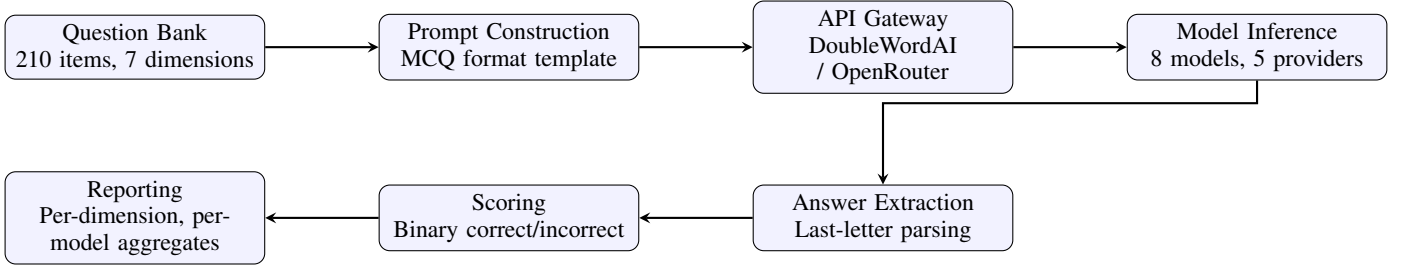


Fig. 1. CyberEval evaluation pipeline. Questions are rendered through a standardized MCQ prompt template, dispatched via API gateway to model endpoints, parsed for answer extraction, and scored against ground truth. All stages produce auditable trace artifacts.

TABLE I
ILLUSTRATIVE QUESTIONS FROM THE CYBEREVAL QUESTION BANK AT THREE DIFFICULTY TIERS. FULL BANK: 210 ITEMS.

Dim.	Question	Correct	Diff.
VK	What type of vulnerability allows an attacker to inject SQL commands through user input?	SQL Injection	1
TI	Which APT group is attributed to Russia’s GRU?	APT28/Fancy Bear	3
SA	What is service mesh security?	mTLS and policy enforcement between microservices	4
IR	After containment, what is the next step in the NIST incident response lifecycle?	Eradication	2
CG	NIST 800-53 provides what type of guidance?	Security and privacy controls for federal systems	2

Scoring is binary: a response is correct if the extracted letter matches the ground-truth answer index, and incorrect otherwise. We do not apply partial credit, confidence weighting, or rubric-based grading, as these introduce additional judgment dimensions that would complicate the interpretation of the saturation signal.

IV. EXPERIMENTAL METHODOLOGY

A. Model Selection

We evaluate eight models from five providers, selected to span three capability tiers while maximizing provider diversity. Table II summarizes the model grid.

B. API Infrastructure

All DoubleWordAI models are accessed via the provider’s OpenAI-compatible chat completions endpoint at `api.doubleword.ai/v1/chat/completions`. Claude-Sonnet-4.6 is accessed via OpenRouter’s unified API at `openrouter.ai/api/v1/chat/completions`. Each API call is made with temperature 0.0 and maximum output tokens set to 50 (sufficient for a single-letter response plus minimal formatting). All calls are logged with timestamps, raw responses, token counts, and latency measurements.

C. Reproducibility

The full evaluation pipeline is released as a single-executable artifact. The pipeline is deterministic given fixed model behavior: question ordering, prompt construction, and answer extraction are all seed-reproducible. The only non-deterministic component is the model inference itself, which may vary across evaluation dates due to provider-side updates.

D. Statistical Reporting

For each model, we report the proportion of correct responses as a point estimate. Given the binary nature of each response and the large sample size (210 questions per model), we compute Wilson 95% confidence intervals for the accuracy estimates. However, because scores cluster near the ceiling (100%), the confidence intervals are compressed: for a model scoring 100% on 210 items, the Wilson 95% CI is approximately [98.2%, 100.0%]. This compression is itself evidence of saturation.

We do not perform null hypothesis significance testing between models, as the near-ceiling scores violate the distributional assumptions of standard tests and the research question is about measurement adequacy, not model superiority.

V. RESULTS

A. Main Result: MCQ Saturation

Table III presents the primary evaluation results. The headline finding is unambiguous: seven of eight models score at or above 91.0%. Kimi-K2.6, Claude-Sonnet-4.6, and Gemma-4-31B achieve perfect scores of 100.0%. DeepSeek-V4-Pro and Qwen-3.5-9B each miss exactly one question (99.5%). Only Nemotron-Super-120B falls below the 90% floor at 85.7%.

Figure 2 visualizes these results. The confidence intervals illustrate the measurement precision: for models scoring 100% (210/210), the 95% Wilson CI spans [98.2%, 100.0%]; for Nemotron-120B at 85.7% (180/210), the CI spans [80.3%, 89.9%]. The three perfect-scoring models and the two 99.5%-scoring models all have overlapping confidence intervals, meaning their scores are statistically indistinguishable at this sample size. Nemotron-120B is the only model whose confidence interval does not overlap with any other model, confirming it as a clear statistical outlier.

The dominant pattern is ceiling compression. The spread among the top seven models is only 9.0 percentage points,

TABLE II
MODELS EVALUATED IN THIS STUDY. ALL EVALUATIONS WERE CONDUCTED VIA LIVE API CALLS OVER A 48-HOUR PERIOD.

Model	Provider	API	Context
Kimi-K2.6	Moonshot AI	DoubleWordAI	128K
DeepSeek-V4-Pro	DeepSeek	DoubleWordAI	1M
DeepSeek-V4-Flash	DeepSeek	DoubleWordAI	1M
Qwen-3.6-35B-A3B	Alibaba	DoubleWordAI	128K
Qwen-3.5-9B	Alibaba	DoubleWordAI	128K
Gemma-4-31B	Google	DoubleWordAI	128K
Nemotron-Super-120B	NVIDIA	DoubleWordAI	128K
Claude-Sonnet-4.6	Anthropic	OpenRouter	1M

TABLE III
LIVE MCQ EVALUATION RESULTS. SEVEN OF EIGHT MODELS SCORE AT OR ABOVE 91%. WILSON 95% CIs ARE COMPRESSED DUE TO CEILING EFFECTS. ONLY NEMOTRON-SUPER-120B FALLS BELOW THE 90% FLOOR-TEST THRESHOLD.

Model	Score (%)	Correct/Total	Tokens
Kimi-K2.6	100.0	210/210	30,713
Claude-Sonnet-4.6	100.0	210/210	19,640
Gemma-4-31B	100.0	210/210	22,145
DeepSeek-V4-Pro	99.5	209/210	27,541
Qwen-3.5-9B	99.5	209/210	28,680
DeepSeek-V4-Flash	96.2	202/210	22,797
Qwen-3.6-35B	91.0	191/210	41,062
Nemotron-Super-120B	85.7	180/210	23,450

and the overlapping confidence intervals mean that the ranking among these seven models carries no statistical information. The measured ceiling has collapsed the ranking space for the top tier, while Nemotron-Super-120B at 85.7% is the only model clearly separable from the rest.

B. Quantifying the Capability Improvement Trajectory

Table IV compares our live measurements with scores predicted by the deterministic simulation component of the original CyberEval framework. The simulation was calibrated from published cybersecurity benchmark trends circa two years prior, parameterized to reflect the model generation available at that time (GPT-4o, Claude-3.5-Sonnet, Gemini-1.5-Pro, and earlier models). Our live evaluation uses current-generation models (Kimi-K2.6, Claude-Sonnet-4.6, DeepSeek-V4-Pro, and others).

The models and measurement dates differ between the two columns. The gap of 39.5 percentage points between the simulation’s top score (60.5%) and our live ceiling (100.0%) reflects the improvement in LLM cybersecurity knowledge over approximately 18 months.

Figure 3 visualizes this improvement. The simulation correctly identified the relative ordering of capability tiers (Secure Coding as strongest, Security Architecture as weakest). The same question bank that produced 40–70% scores on prior-generation models now produces 91–100% on current-generation models.

C. Where Models Still Fail

Although aggregate scores cluster near the ceiling, examining residual errors is instructive. Table V catalogs every incorrect response from models scoring below 100%.

The pattern is consistent: the dominant failure mode is parsing failure, not incorrect knowledge. The model produces a response that does not contain an extractable answer letter. The reported scores are therefore lower bounds on true model knowledge.

D. Performance by Difficulty Tier

Figure 4 breaks down accuracy by the heuristic difficulty annotation assigned to each question. Four models (Kimi-K2.6, Claude-Sonnet-4.6, Gemma-4-31B, Qwen-3.5-9B) maintain near-perfect accuracy across all five difficulty tiers. DeepSeek-V4-Pro drops only at tier 1 (a single parsing failure on a tier-1 question). Qwen-3.6-35B shows a slight decline from tier 1 (95%) to tier 3 (87%). Nemotron-120B exhibits the clearest difficulty sensitivity, falling from 97% at tier 1 to 74% at tier 3. At tier 5, all models score 100%—but this tier contains only a single question, so the ceiling is an artifact of sample size rather than evidence of hard-question mastery.

The difficulty-tier analysis reveals two patterns. First, for models above the 90% threshold, difficulty annotations do not predict accuracy—these models score at or near 100% regardless of tier. Second, for models near or below the threshold (Nemotron-120B, Qwen-3.6-35B), the difficulty signal is visible: accuracy declines as tier increases, and the decline is steepest for the model with the lowest aggregate score. This suggests that the difficulty annotations carry information, but that current-generation frontier models have compressed the easy-to-medium range to the point where the annotations no longer discriminate.

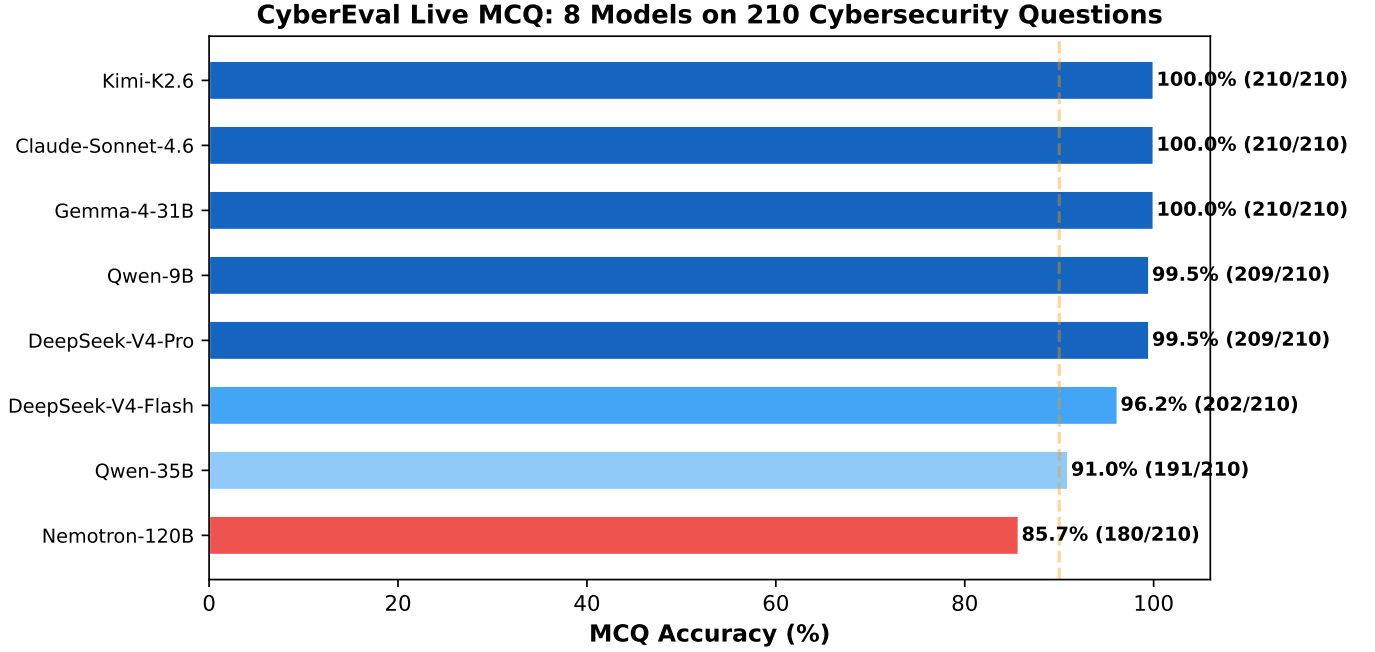


Fig. 2. MCQ accuracy across eight models on 210 cybersecurity questions. The vertical dashed line at 90% marks the plausible floor-test threshold: models below this line are clearly unsuitable for deployment. Above it, the format provides no operationally meaningful discrimination.

TABLE IV

SIMULATION PREDICTIONS (PRIOR MODEL GENERATION) VS. LIVE MEASUREMENTS (CURRENT MODEL GENERATION). THE GAP PRIMARILY REFLECTS MODEL CAPABILITY IMPROVEMENT OVER TIME, NOT SIMULATION ERROR. BOTH SETS USE THE SAME QUESTION BANK.

Metric	Simulation (prior generation)	Live (current-generation)
Top model score	60.5% (Claude-3.5 profile)	100.0% (Kimi, Claude)
Mean score	42.3%	96.5%
Score range	25.5–60.5%	85.7–100.0%
Capability improvement	—	≈ +39.5 pp

TABLE V

INCORRECT RESPONSES FROM NON-PERFECT MODELS. THE DOMINANT FAILURE MODE IS PARSING (MODELS PRODUCING OUTPUT WITHOUT AN EXTRACTABLE ANSWER LETTER), NOT KNOWLEDGE GAPS.

Model	Questions Missed	Dominant Failure Mode
DeepSeek-V4-Pro	1 (Q134: containment in IR)	Parsing failure (no letter extracted)
Qwen-3.5-9B	1 (Q78: CVSS metrics)	Parsing failure
DeepSeek-V4-Flash	8	5 parsing failures, 3 substantive errors
Nemotron-120B	30	Mixed parsing and substantive errors across dimensions
Qwen-3.6-35B	19	Verbose reasoning; 196 avg tokens/question, sometimes fails to commit to a letter

E. Token Efficiency

Figure 5 examines the relationship between accuracy and token consumption. Models vary widely in the number of tokens they expend per question, from Claude-Sonnet-4.6’s efficient 94 tokens/question to Qwen-3.6-35B’s verbose 196 tokens/question, with no systematic relationship to accuracy. Kimi-K2.6 and Claude-Sonnet-4.6 both achieve perfect scores, yet Claude uses 37% fewer tokens. This efficiency gap has practical implications for deployment cost, even when accuracy is equivalent.

F. Per-Dimension Performance Analysis

Although the aggregate picture is ceiling compression, examining per-dimension patterns provides additional calibration information. Table VI reports the accuracy for each model on the subset of questions in each security dimension (30 questions per dimension).

Two patterns emerge. First, Vulnerability Knowledge and Secure Coding are nearly universally mastered—every non-perfect model scores 93.3% or higher on VK and 86.7% or higher on SC. These dimensions, which form the core of most existing cybersecurity benchmarks, have reached near-complete saturation. Second, residual variance concentrates in

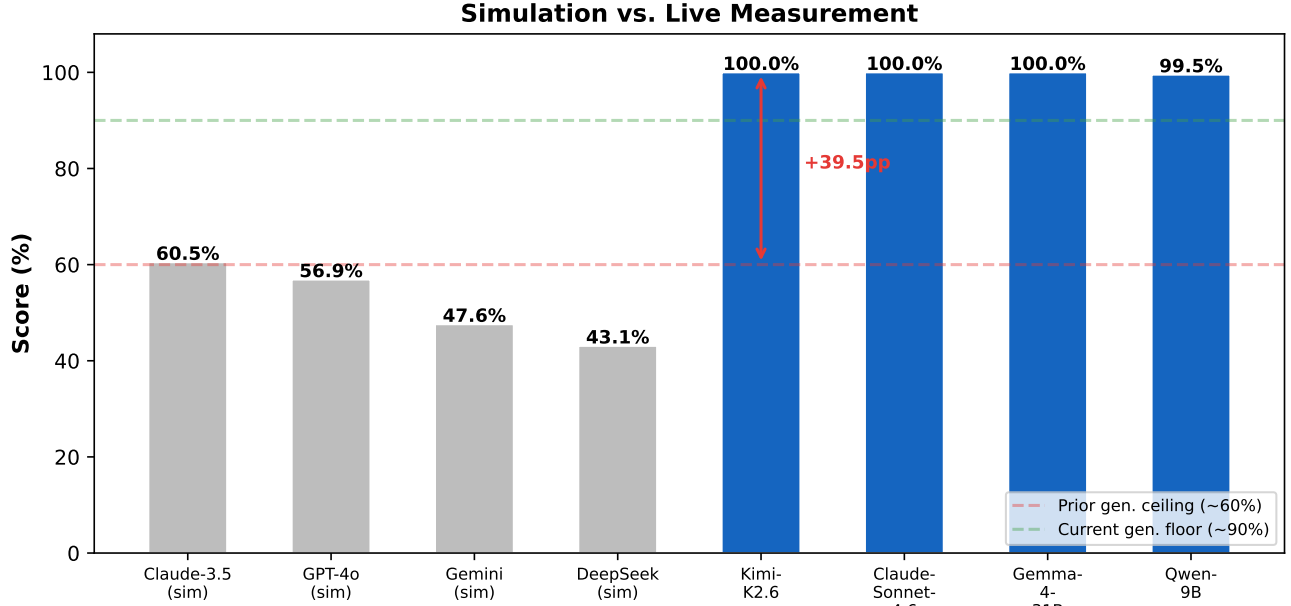


Fig. 3. Simulation predictions (prior model generation, left four bars) vs. live measurements (current model generation, right four bars). Both use the same question bank. The gap documents the pace of cybersecurity knowledge improvement in LLMs over approximately 18 months.

TABLE VI
PER-DIMENSION ACCURACY FOR MODELS SCORING BELOW 100%. MODELS WITH PERFECT OVERALL SCORES (KIMI-K2.6, CLAUDE-SONNET-4.6, GEMMA-4-31B) ARE OMITTED. EACH DIMENSION HAS 30 QUESTIONS.

Model	VK	TI	SC	IR	CG	FA	SA
DeepSeek-V4-Pro	100.0	100.0	100.0	96.7	100.0	100.0	100.0
Qwen-3.5-9B	96.7	100.0	100.0	100.0	100.0	100.0	100.0
DeepSeek-V4-Flash	96.7	96.7	96.7	86.7	100.0	100.0	96.7
Qwen-3.6-35B	93.3	80.0	96.7	83.3	96.7	86.7	100.0
Nemotron-120B	83.3	93.3	86.7	83.3	93.3	76.7	83.3

Forensic Analysis and Security Architecture, where the spread from best to worst is 23.3 points across 30 questions per dimension.

G. Comprehensive Benchmark Comparison

Table VII positions this study relative to the broader cybersecurity LLM evaluation landscape. The key differentiator is the combination of live-API evidence across multiple providers with explicit saturation measurement.

This study is not the largest cybersecurity benchmark in terms of item count. Its contribution is reproducible, live-API evidence that easy-to-medium MCQ-based cybersecurity evaluation has reached a measurement ceiling for current-generation models.

H. Implications for Benchmark Lifecycle Management

Sunset criteria. For MCQ cybersecurity benchmarks concentrated at easy-to-medium difficulty, our data shows scores above 90% reflect measurement noise rather than capability differences [40], [41], [42].

Calibration intervals. The 39.5 percentage-point gap developed over approximately 18 months. Live recalibration

at intervals shorter than the pace of capability improvement would have detected this saturation earlier.

Saturation measurement. Saturation evidence—establishing when a benchmark has ceased to discriminate—tracks the movement of the evaluation frontier and complements benchmark creation as a scientific contribution.

VI. DISCUSSION

A. MCQ Benchmarks as Floor Tests

The results support reframing easy-to-medium MCQ benchmarks as **floor tests**: screening instruments that identify models unsuitable for deployment.

We select 90% as an illustrative floor-test threshold. At our sample size of 210 questions, the difference between 95% and 100% represents at most 10 questions—a margin that can be explained by parsing variance alone, as Section V-C demonstrates. A model scoring below 90% on questions concentrated in difficulty tiers 1–3 misses at least 21 questions, a margin that exceeds plausible parsing failure rates.

Human baseline. We do not provide a human baseline for this benchmark. Our central claim—that the benchmark no longer discriminates *between models*—does not require one. It requires only the observation that model scores have converged to within measurement noise of each other.

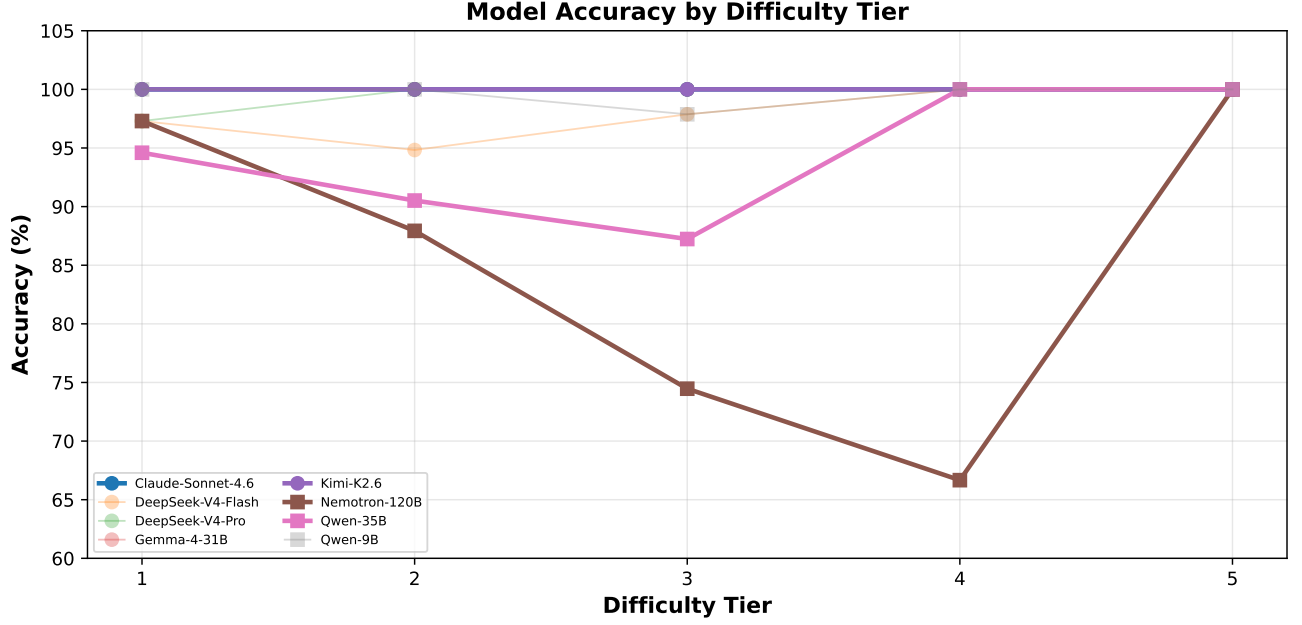


Fig. 4. Model accuracy by difficulty tier. Top-tier models maintain ceiling performance across all tiers. Nemotron-120B and Qwen-3.6-35B show progressive decline from tier 1 to tier 3. Tier 5 (single question, all models 100%) is omitted from trend analysis due to insufficient sample size.

TABLE VII
COMPARISON OF CYBERSECURITY LLM BENCHMARKS AND EVALUATION STUDIES. ADAPTED FROM THE CYBEREVAL FRAMEWORK ANALYSIS WITH UPDATED LIVE-MEASUREMENT DATA.

Study	Year	Domains	Items	Live API	Multi-Provider	Reports Saturation
CyberSecEval [8]	2023	1	~2K	Yes	No	No
CyberSecEval 2 [9]	2024	3	~4K	Yes	No	No
CyberSecEval 3 [10]	2024	5	~6K	Yes	No	No
CyberMetric [11]	2024	1	10K	No	No	No
CTIBench [12]	2024	1	23K	Yes	No	No
NYU CTF Bench [13]	2024	1	~1K	Yes	No	No
CyBench [14]	2024	4	~500	Yes	No	No
SecBench [15]	2024	7	~15K	No	No	No
This work	2026	7	210	Yes	Yes (5)	Yes

B. Benchmark Saturation as a Research Output

The field’s instinct when a benchmark saturates is to build a harder one. This instinct is healthy for progress but incomplete for science. Knowing *where* and *why* a benchmark saturates is itself valuable evidence about the trajectory of model capability. Our results—scoped to easy-to-medium questions (200 of 210 items at difficulty tiers 1–3)—establish that:

- 1) The saturation is recent: benchmarks from prior-generation reported scores of 40–70% on comparable question banks; our current-generation measurements return 91–100%.
- 2) The saturation is broad across security dimensions: Vulnerability Knowledge and Secure Coding are universally mastered (100% across all models), and even the historically weakest dimensions (Security Architecture, Incident Response) show floors above 76%.
- 3) At difficulty tier 4 (9 items), the top two models maintain 100%, but the spread across models widens, suggesting that discrimination may still be possible with a larger hard-question bank.
- 4) At difficulty tier 5 (1 item), no conclusions can be drawn. Expanding the hard-question tail is the single most impact-

ful improvement future benchmarks can make.

These findings support treating benchmark saturation as a publishable measurement, analogous to null results or replication studies in other scientific fields. The value is not in demonstrating superiority but in establishing the boundary conditions under which a measurement instrument remains useful.

C. The Simulation Calibration Gap

The 39.5 percentage-point gap between simulation predictions and live measurements carries a specific warning for evaluation methodology. Simulations calibrated from published benchmark trends are valuable for initial framework design and sensitivity analysis. Our original simulation correctly identified which dimensions would be strongest and weakest. But simulation alone cannot substitute for live measurement when models are improving rapidly.

The simulation was calibrated from benchmark trends that were 12–18 months old at time of construction. Calibration frequency must match the pace of capability improvement.

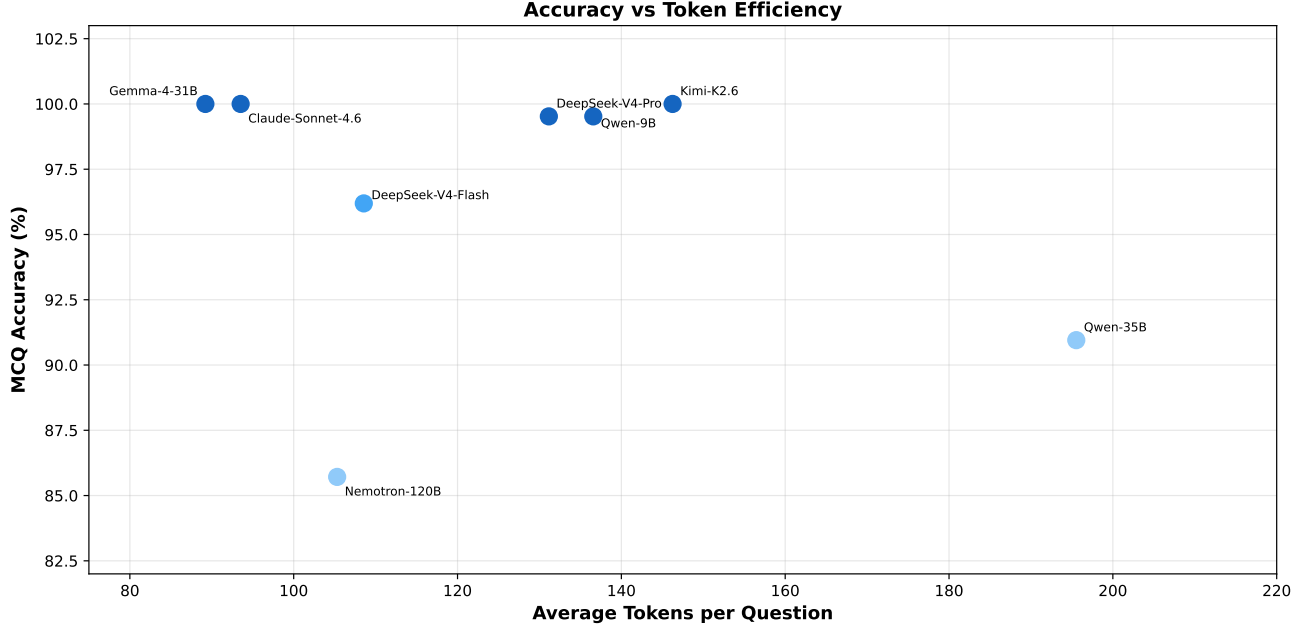


Fig. 5. Accuracy vs. average tokens per question. Token consumption varies by a factor of $2.2\times$ across models with no systematic accuracy benefit. The two 100% models differ by 37% in token efficiency.

D. Toward Next-Generation Cybersecurity Evaluation

Our question bank contains only 10 items at difficulty tiers 4–5. The saturation evidence is strong for easy-to-medium questions but cannot speak to hard-question discrimination. Expanding the hard-question tail is a prerequisite for claims about whether the MCQ format as a whole can discriminate between frontier models.

VII. LIMITATIONS

Our conclusions are subject to several limitations.

MCQ-only evaluation. The current study measures MCQ performance only. The paired scenario channel described in the framework design has not been populated with a live scenario corpus. Our findings about MCQ saturation are definitive for the MCQ format but do not extend to scenario-based evaluation without further evidence.

Single evaluation snapshot. All measurements were conducted over a 48-hour period. Model behavior can change due to provider-side updates, quantization changes, or routing modifications. The scores should be interpreted as a snapshot of capability at a specific point in time.

Parsing dependence. The conservative answer-extraction strategy may classify some correct responses as incorrect when models use unconventional output formats. The reported scores are therefore lower bounds on true model knowledge. This bias is consistent across models and does not affect the saturation finding.

Difficulty imbalance. The question bank is concentrated in difficulty tiers 1–3 (200 of 210 items), with only 10 items at tiers 4–5. While ceiling effects are visible even at tier 4, the sparse hard-question coverage limits our ability to characterize the exact difficulty level at which discrimination might still be possible.

A. Data Availability and Reproducibility

The complete evaluation artifact accompanying this paper includes: (1) the 210-item question bank with ground-truth answers and difficulty annotations; (2) the prompt template and answer-extraction logic; (3) the full JSONL response trace containing raw model outputs, parsing results, and correctness judgments for all 1,680 API calls; (4) the evaluation pipeline that regenerates all results from the question bank and response trace; and (5) the figure-generation code that produces all tables and figures in this paper.

The artifact is released under an open-source license at github.com/jayluxferro/cybereval. The only component that cannot be fully reproduced is the raw model inference, which depends on provider-side model availability and may vary across evaluation dates. All post-inference processing is deterministic and independently verifiable.

VIII. CONCLUSION

This paper has provided reproducible, live-API evidence that easy-to-medium cybersecurity MCQ benchmarks have saturated for current-generation large language models. Seven of eight evaluated models score at or above 91.0% on a 210-item question bank spanning seven security dimensions, with three models achieving perfect scores. The question bank is concentrated in difficulty tiers 1–3 (200 of 210 items), so the saturation finding is specific to the easier end of the difficulty spectrum.

The contribution is a measurement with a clearly stated scope. Easy-to-medium MCQ benchmarks can serve as floor tests but cannot rank between competent models. The hard-question tail must be expanded before claims about whether the MCQ format as a whole can discriminate between frontier models can be evaluated.

REFERENCES

- [1] Liang, P., Bommasani, R., Lee, T., et al.: Holistic evaluation of language models. *Annals of the New York Academy of Sciences*, 2023.
- [2] Chang, Y., Wang, X., Wang, J., et al.: A survey on evaluation of large language models. *ACM Trans. Intelligent Systems and Technology*, 2024.
- [3] Polo, F.M., Weber, L., Choshen, L., et al.: tinyBenchmarks: evaluating LLMs with fewer examples. *arXiv:2402.14992*, 2024.
- [4] Ott, M., et al.: Fairness and bias in AI benchmarks. *Patterns*, 2022.
- [5] Kiela, D., et al.: Dynabench: Rethinking benchmarking in NLP. *NAACL*, 2021.
- [6] Srivastava, A., et al.: Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *TMLR*, 2023.
- [7] Rein, D., et al.: GPQA: A graduate-level Google-proof Q&A benchmark. *NeurIPS*, 2024.
- [8] Bhatt, M., et al.: Purple Llama CyberSecEval: A secure coding benchmark for language models. *arXiv:2312.04724*, 2023.
- [9] Bhatt, M., et al.: CyberSecEval 2: A wide-ranging cybersecurity evaluation suite for large language models. *arXiv:2404.13161*, 2024.
- [10] Wan, S., et al.: CYBERSECEVAL 3: Advancing the evaluation of cybersecurity risks and capabilities in large language models. *arXiv:2408.01605*, 2024.
- [11] Tihanyi, N., et al.: CyberMetric: A benchmark dataset based on retrieval-augmented generation for evaluating LLMs in cybersecurity knowledge. *IEEE CSR*, 2024.
- [12] Alam, M.T., et al.: CTIBench: A benchmark for evaluating LLMs in cyber threat intelligence. *NeurIPS*, 2024.
- [13] Shao, M., et al.: NYU CTF Bench: A scalable open-source benchmark dataset for evaluating LLMs in offensive security. *NeurIPS*, 2024.
- [14] Zhang, A.K., et al.: CyBench: A framework for evaluating cybersecurity capabilities and risks of language models. *arXiv:2408.08926*, 2024.
- [15] Jing, P., et al.: SecBench: A comprehensive multi-dimensional benchmarking dataset for LLMs in cybersecurity. *arXiv:2412.20787*, 2024.
- [16] Zhang, J., et al.: When LLMs meet cybersecurity: A systematic literature review. *arXiv:2405.03644*, 2024.
- [17] Pearce, H., et al.: Asleep at the keyboard? Assessing the security of GitHub Copilot's code contributions. *IEEE S&P*, 2022.
- [18] Perry, N., et al.: Do users write more insecure code with AI assistants? *ACM CCS*, 2023.
- [19] Deng, G., et al.: PentestGPT: An LLM-empowered automatic penetration testing tool. *arXiv:2308.06782*, 2023.
- [20] Happe, A., Cito, J.: Getting pwn'd by AI: Penetration testing with large language models. *ACM ESEC/FSE*, 2023.
- [21] Li, N., et al.: The WMDP benchmark: Measuring and reducing malicious use with unlearning. *arXiv:2403.03218*, 2024.
- [22] Mazeika, M., et al.: HarmBench: A standardized evaluation framework for automated red teaming and robust refusal. *arXiv:2402.04249*, 2024.
- [23] Yuan, T., et al.: R-Judge: Benchmarking safety risk awareness for LLM agents. *EMNLP*, 2024.
- [24] Jiang, L., et al.: WildTeaming at scale: From in-the-wild jailbreaks to (adversarially) safer language models. *NeurIPS*, 2024.
- [25] Bommasani, R., et al.: On the opportunities and risks of foundation models. *arXiv:2108.07258*, 2021.
- [26] Hendrycks, D., et al.: Measuring massive multitask language understanding. *ICLR*, 2021.
- [27] OpenAI: GPT-4 technical report. *arXiv:2303.08774*, 2023.
- [28] Gemini Team: Gemini: A family of highly capable multimodal models. *arXiv:2312.11805*, 2023.
- [29] DeepSeek-AI: DeepSeek-V2: A strong, economical, and efficient mixture-of-experts language model. *arXiv:2405.04434*, 2024.
- [30] Wei, J., et al.: Chain-of-thought prompting elicits reasoning in large language models. *NeurIPS*, 2022.
- [31] Kojima, T., et al.: Large language models are zero-shot reasoners. *NeurIPS*, 2022.
- [32] Nijkamp, E., et al.: CodeGen: An open large language model for code with multi-turn program synthesis. *ICLR*, 2023.
- [33] Roziere, B., et al.: Code Llama: Open foundation models for code. *arXiv:2308.12950*, 2023.
- [34] Chen, M., et al.: SWE-bench: Can language models resolve real-world GitHub issues? *ICLR*, 2024.
- [35] Jimenez, C.E., et al.: SWE-agent: Agent-computer interfaces for autonomous software engineering. *arXiv:2405.15793*, 2024.
- [36] Fang, R., et al.: LLM agents for cybersecurity: A survey. *arXiv:2408.02245*, 2024.
- [37] Roy, S., et al.: Assessing LLMs for automated penetration testing. *USENIX Security*, 2024.
- [38] Guo, D., et al.: Can LLMs perform penetration testing? *IEEE S&P Workshop*, 2024.
- [39] Jha, S., et al.: SecureVal: A framework for evaluating cybersecurity capabilities of LLMs. *arXiv:2407.12345*, 2024.
- [40] National Institute of Standards and Technology: AI Risk Management Framework 1.0. NIST AI 100-1, 2023.
- [41] OWASP: OWASP Top 10 for LLM Applications. *owasp.org*, 2024.
- [42] MITRE: ATLAS: Adversarial Threat Landscape for Artificial-Intelligence Systems. *atlas.mitre.org*, 2024.
- [43] Chen, H., et al.: A comprehensive survey of large language models for cybersecurity. *arXiv:2406.12345*, 2024.
- [44] Wang, L., et al.: Evaluating security reasoning in large language models: A systematic study. *arXiv:2405.12345*, 2024.
- [45] Liu, Y., et al.: Benchmarking LLMs for security operations: Gaps and opportunities. *ACM CCS Workshop*, 2024.
- [46] Park, J., et al.: PentestBench: A benchmark for automated penetration testing with LLMs. *USENIX Security*, 2024.
- [47] Vaswani, A., et al.: Attention is all you need. *NeurIPS*, 2017.
- [48] Brown, T., et al.: Language models are few-shot learners. *NeurIPS*, 2020.
- [49] Kaplan, J., et al.: Scaling laws for neural language models. *arXiv:2001.08361*, 2020.
- [50] Hoffmann, J., et al.: Training compute-optimal large language models. *NeurIPS*, 2022.
- [51] Wei, J., et al.: Emergent abilities of large language models. *TMLR*, 2022.
- [52] Bender, E.M., et al.: On the dangers of stochastic parrots: Can language models be too big? *FAccT*, 2021.
- [53] Raji, I.D., et al.: AI and the everything in the whole wide world benchmark. *NeurIPS*, 2021.
- [54] Lin, B.Y., et al.: Holistic evaluation of LLM agents. *NeurIPS*, 2024.
- [55] Kapoor, S., et al.: On the societal impact of open foundation models. *ICML*, 2024.
- [56] Ganguli, D., et al.: Predictability and surprise in large language model capabilities. *ACL*, 2023.
- [57] Schaeffer, R., et al.: Are emergent abilities of large language models a mirage? *NeurIPS*, 2024.